Xue-Qiang Fan, Ph.D.

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About Me

I am currently a Ph.D. Candidate major in School of Computer and Information, Information and Communication Engineering, Hefei University of Technology.

- **Interests:** My research interests include Artificial Intelligence, Machine Learning, and Data Mining. In particular, I am interested in Pattern Recognition, Polarization Vision, Computational Imaging, and Bioinformatics.
- **Address:** Danxia Road 485, Shushan District, Hefei, Anhui, China.

Education

Jun. 2022 – Present
Ph.D. Hefei University of Technology, Hefei, China
Major: *Information and Communication Engineering, School of Computer and Information.* ***Advisor: Prof. Zhong-Yi Guo***

Jul. 2019 – May. 2022
M.S. Zhejiang University of Technology, Hangzhou, China
Major: *Control Science and Engineering, College of Information Engineering.* ***Advisor: Prof. Jun Hu***.

Sep. 2015 - Jun. 2019
B.S. Henan University of Urban Construction, Pingdingshan, China
Major: *Automation, School of Electrical and Control Engineering. Courses: Automatic Control Theory, etc..*

Research Experience

Zhong-Yi Guo Research Group, Advanced Electromagnetic Function Laboratory (AEMFLab)

June. 2022 – Present
Interests: Pattern Recognition, Polarization Vision, Computational Imaging.

Gui-Jun Zhang Research Group, Intelligent Optimization and Bioinformatics Lab (IOBIO-Lab)

July. 2019 - May. 2022
Interests: Pattern Recognition, Bioinformatics.

Research Publications

(# Contribution equally; * Corresponding authors)

Journal Articles (Featured Publications)

1. **X. Q. Fan**, W. Y. Chen, B. Lin, P. Peng, and Z. Y. Guo*, "Improved polarization scattering imaging using local-global context polarization feature learning framework," *Opt. Lasers Eng.*, vol. 178, p. 108 194, 2024. DOI: 10.1016/j.optlaseng.2024.108194.
2. **X. Q. Fan**, B. Lin, K. Guo, B. Y. Liu, and Z. Y. Guo*, "TSMPN-PSI: High-performance polarization scattering imaging based on three-stage multi-pipeline networks," *Opt. Express*, vol. 31, pp. 38 097–38 113, 2024. DOI: 10.1364/OE.501269.
3. **X. Q. Fan**, B. Lin, and Z. Y. Guo*, "Infrared polarization-empowered full-time road detection via lightweight multi-pathway collaborative 2d/3d convolutional networks," *IEEE Trans. Intell. Transp. Syst. In Press*, 2024. DOI: 10.1109/TITS.2024.3383405.

4. B. Lin, **X. Q. Fan**, P. Peng, and Z. Y. Guo*, "Dynamic polarization fusion network (DPFN) for imaging in different scattering systems," *Opt. Express*, vol. 32, pp. 511–525, 2024. DOI: 10.1364/OE.507711.

5. **X. Q. Fan**, B. Lin, J. Hu, and Z. Y. Guo*, "I-DNAN6mA: Accurate identification of dna n6-methyladenine sites using the base-pairing map and deep learning," *J. Chem. Inf. Model.*, vol. 63, no. 3, pp. 1076–1086, 2023. DOI: 10.1021/acs.jcim.2c01465.

6. B. Lin, **X. Q. Fan**, and Z. Y. Guo*, "Self-attention module in multi-scale improved u-net (SAM-MIU-net) motivating high-performance polarization scattering imaging," *Opt. Express*, vol. 31, no. 2, pp. 3046–3058, 2023. DOI: 10.1364/OE.479636.

7. **X. Q. Fan**, J. Hu*, Y. X. Tang, D. J. Yu*, and G. J. Zhang*, "Predicting rna solvent accessibility from multi-scale context feature via multi-shot neural network," *Analytical Biochemistry*, vol. 654, p. 114 802, 2022. DOI: 10.1016/j.ab.2022.114802.

8. **X. Q. Fan**, J. Hu*, N. X. Jia, D. J. Yu*, and G. J. Zhang*, "Improved protein relative solvent accessibility prediction using deep multi-view feature learning framework," *Analytical Biochemistry*, vol. 631, p. 114 358, 2021. DOI: 10.1016/10.1016/j.ab.2021.114358.

Miscellaneous Experience

Awards and Achievements

2021 **The National Scholarship**, Awarded by Zhejiang University of Technology.

2020 **The National Scholarship**, Awarded by Zhejiang University of Technology.

Skills

Programming Languages	Python, Linux Shell Script, Matlab, Java, LaTeX, …
Misc.	Academic research, LaTeX typesetting, …
Hobbies	Running, Road Cycling, Folk Music, Movie, …